

# Solar Photovoltaic Devices as Radiation Sensors for Post-detonation Nuclear Forensics

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**Abstract**—In this study, we evaluated the feasibility of applying solar photovoltaic (PV) panels as sensors of nuclear and electromagnetic radiation that includes neutrons, x-rays and gamma-rays, and optical radiation emanating from a nuclear explosion. We investigated the steady-state and transient response of both a commercial silicon (Si) and a perovskite solar cell to different radiation types. Solar cell current-voltage characteristics and short-circuit current ( $I_{sc}$ ) response under steady-state x-ray illumination were measured. The fast transient radiation pulse from a nuclear detonation was mimicked by using a fast switching, nanosecond pulsed laser source and the transient response of the solar cells was captured on an oscilloscope. Subsequently, the transient response of Si solar cells to pulsed x-rays generated by a mechanical x-ray chopper was measured. A 2 MeV neutron beam chopper was built at the fast neutron beam facility of a research reactor to produce time-modulated neutrons and evaluate the solar cell transient response to a neutron pulse. Our steady-state measurements demonstrated good response of solar cells to x-rays and neutrons. The pulsed radiation measurements indicated that the solar cells are able to detect a fast transient radiation and produce a proportional measurable output signature.

## I. INTRODUCTION

NUCLEAR explosions involve the release of enormous amount of energy, typically expressed as equivalent to thousands of tons of a TNT explosive. Energy from a nuclear explosion is emitted in a complex manner in various forms that includes an intense spectrum of electromagnetic radiation comprising prompt gamma-rays from fission, neutron capture gamma-rays, x-rays from thermal radiation, visible light and radio waves [1]. Additionally, the initial nuclear radiation consists of short-ranged prompt neutrons from nuclear fission, whereas the residual radiation consists of delayed neutrons from fission product decay and those spontaneous-fission neutrons from radioactive fallout. A significant amount of

energy is carried away by air blast/shock wave and includes infrasonic, seismic and air pressure waves. The different forms of energy release point to different practical modes of detection of a nuclear explosion – through infrasonic, seismic and hydroacoustic sensing, and thermal radiation sensors etc. Sensor networks and systems corresponding to such detection methods are currently deployed at suitable locations across the world [2][3]. Although such sensor systems provide useful data in response to external triggers from a nuclear explosion, there are several possible scenarios where the sensor data may prove insufficient to promptly identify a nuclear detonation and provide forensic information. For example, in surface or atmospheric nuclear explosions, there is potentially high interference to the sensor signal from that induced by earthquakes.

Solar photovoltaic (PV) panels are now ubiquitously deployed as urban manmade infrastructure, and they can serve as potential sensors to detect nuclear detonation, providing information both complementary and supplementary to that from other sensor systems. A single solar PV panel/array might not be sensitive enough to detect the radiative output of a nuclear detonation, but a solar farm or a network of solar arrays can produce a signature change in power output from baseline before they are destroyed by the electromagnetic pulse, the thermal radiation and/or the shock wave. In addition, the solar panels that survive the explosion may exhibit delayed measurable signatures from radiation interactions with its active components, thereby providing both prompt and delayed post-detonation forensics information.

What makes the solar panel or a solar array to detect a nuclear explosion is the active p-n junction that is sensitive to nuclear radiation, as it does to light. Silicon (Si), the most common material used in commercial solar cell development, is also sensitive to fast neutrons due to neutron elastic scattering and other energy threshold reactions. In an early study in 1960, Si solar cells were already evaluated as x- and gamma-ray dosimeters and radiation monitors [4]. More recently, commercial low power Si PV cells were evaluated for gamma-ray dosimetry in space [5]. In a previous study [6], the energy deposition spectrum of reactor neutrons in a SiC detector at a near-core location of a research reactor was measured. The experimental spectrum was compared against a simulation result of 2 MeV neutron energy deposition spectrum in Si, and the sensitivity of Si to fast neutrons was demonstrated [6].

Perovskite solar cells have recently emerged as promising thin film PV technology [7], with high power conversion efficiency of perovskite solar cells already demonstrated by

Manuscript received December 15, 2020. This work was supported by the U.S. Department of Defense, Defense Threat Reduction Agency under Grant No. HDTRA11910024.

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various research groups. Due to the high effective atomic number and hydrogenous medium of organic lead halide perovskite (e.g.  $\text{CH}_3\text{NH}_3\text{PbI}_3$ ), it is inherently sensitive to both gamma-rays and fast neutrons.

In this study, we evaluated both Si and perovskite solar cells and quantified their response to ionizing radiation.

## II. METHODS

### A. Operation of a Solar Cell

The operation of a solar cell is similar to that of a semiconductor radiation detector operated at zero bias voltage. The current generated in the solar cells of a solar PV panel can be used to drive an external load. To evaluate a PV panel response to prompt nuclear radiation, it is imperative to have a fundamental understanding of the physics of radiation interactions and energy deposition, charge carrier generation and recombination, and carrier transport through drift and diffusion in solar cell. With such knowledge, a solar cell, at circuit level, can be represented as a photogenerated current source in parallel with a p-n junction diode. The photogenerated current,  $I_{ph}$  of a solar cell is the current induced in it by visible light or ionizing radiation. For an ideal solar cell, the short-circuit current,  $I_{sc}$  equals  $I_{ph}$ . However, in practice,  $I_{sc}$  is lower than  $I_{ph}$  due to losses introduced by solar cell series and shunt resistances [8]. Therefore, a more accurate circuit-level representation of solar cell may be as shown in Fig. 1, which also includes the capacitance to account for the dynamic behavior of solar cell along with its static response [9].

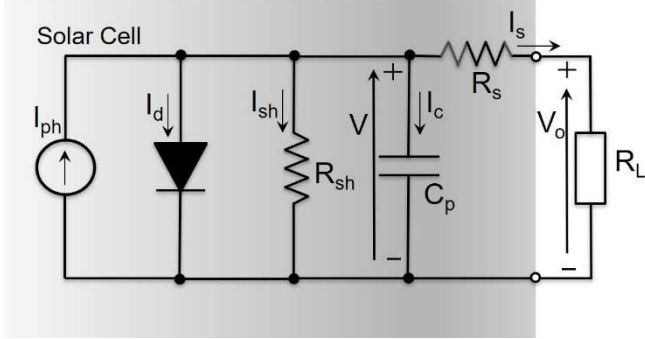


Fig. 1. Equivalent circuit model of a photovoltaic solar cell.

In Fig. 1,  $I_{ph}$  is the photogenerated current;  $I_d$  is the diode current;  $I_{sh}$  is the current flowing through shunt resistor;  $I_c$  is the current flowing through capacitor;  $I_s$  is the output current driving the load;  $R_s$  is the solar cell series resistance;  $R_{sh}$  is the solar cell shunt resistance;  $R_L$  is the load resistance; and  $C_p$  is the solar cell capacitance. The relationship between  $I_s$  and  $I_{ph}$  is expressed as:

$$I_s = I_{ph} - I_d - I_{sh} - I_c; \text{ where}$$

$I_d = I_0 \left( e^{\frac{qV}{\eta kT}} - 1 \right)$ ; where  $I_0$  is the diode saturation current,  $q$  is the electron elementary charge,  $\eta$  is the diode ideality factor,  $k$  is the Boltzmann constant,  $T$  is the diode temperature, and  $V$  is the voltage across the diode.

$$I_{sh} = \frac{V}{R_{sh}};$$

$$I_c = C_p \frac{dV}{dt}; \text{ and } I_s = \frac{V}{R_s + R_L} = \frac{V_o}{R_L}. \text{ Thus,}$$

$$I_s = I_{ph} - I_0 \left( e^{\frac{qI_s(R_s + R_L)}{\eta kT}} - 1 \right) - \frac{I_s(R_s + R_L)}{R_{sh}} - C_p \frac{dI_s(R_s + R_L)}{dt}$$

At zero load, the output voltage  $V_o = 0$  as  $R_L = 0$ , and  $I_s = I_{sc}$ . At infinite load, the output current  $I_s = 0$  as  $R_L = \infty$ , and  $V = V_o = V_{oc}$ , i.e., open-circuit voltage.

In the current study, we experimentally evaluated the solar cell response to ionizing radiation by measuring the solar cell output current  $I_s (= I_{sc}$ , when  $R_L = 0$ ) and output voltage  $V_o (= V_{oc}$ , when  $R_L = \infty$ ). For these measurements, we used a commercially purchased polycrystalline Si (Kitronik, 3604) and in-house fabricated perovskite solar cells. The Si solar cell has about 185  $\mu\text{m}$  thick n<sup>+</sup>-p Si layer. Electron Microscopy images shown in Fig. 2(a) indicate the different layers including the optically transparent protective epoxy layer in the Si solar cell. Fig. 2(b) shows a schematic of the thin-film methyl ammonium lead triiodide perovskite solar cell. The perovskite solar cell modules were mounted on a breadboard to enable measurements (Fig. 2(c)).

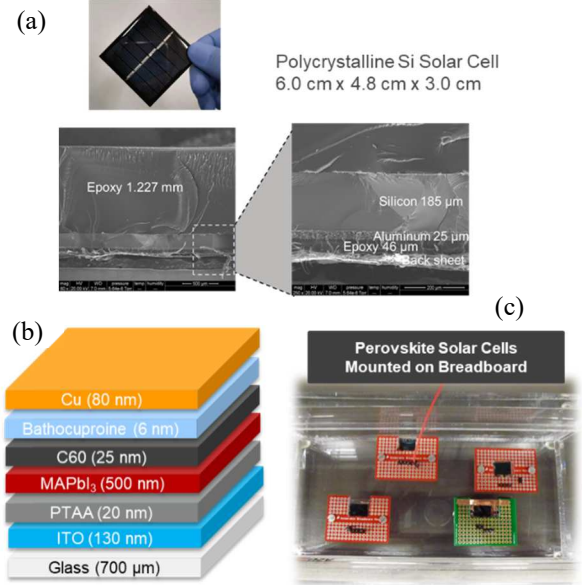


Fig. 2. (a) Polycrystalline Si solar cell (Kitronik, 3604) and electron microscopy images revealing the structure of Si solar cell (b) A schematic of the structure of in-house fabricated perovskite solar cell (c) Perovskite solar modules mounted on circuit boards prior to the measurements

### B. Steady State X-ray Response

At first the steady-state response of solar cells to x-rays was measured to evaluate their sensitivity to short wavelength photons. The solar cells were placed in a dark chamber and exposed to x-rays from a miniature stand-alone x-ray tube connected to the chamber (Fig. 3(a)). Specifications of the x-ray source are summarized in Table I. The solar cell I-V characteristics and short-circuit current response under x-ray illumination at various x-ray tube voltages (10 kV to 50 kV) were measured using a semiconductor parameter analyzer (Keithley 4200A-SCS) (Fig. 3(b)).

TABLE I. SPECIFICATIONS OF THE X-RAY SOURCE USED IN THIS STUDY

|                        |                                      |
|------------------------|--------------------------------------|
| Manufacturer, Model    | Amptek, Mini-X X-ray Tube System     |
| X-ray Tube Design      | Transmission type                    |
| Target Material        | Tungsten                             |
| Target Thickness       | 0.75 $\mu\text{m}$                   |
| Window Material        | Beryllium                            |
| Window Thickness       | 125 $\mu\text{m}$                    |
| Tube Operating Voltage | 10 kV to 50 kV                       |
| Tube Operating Current | 5 $\mu\text{A}$ to 200 $\mu\text{A}$ |
| Tube Output Power      | 4 W (max.)                           |

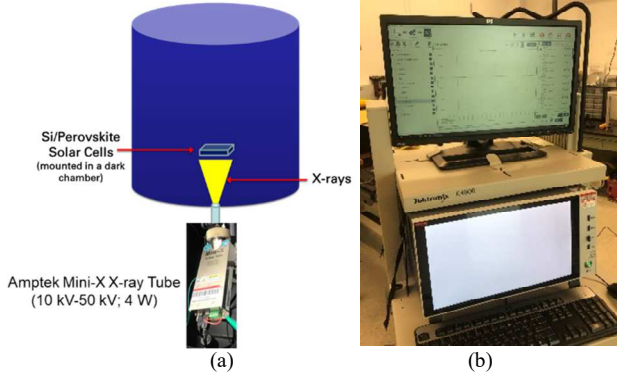


Fig. 3. (a) An illustration of the measurement setup used in solar cell steady-state x-ray measurements, showing the miniature x-ray tube (Amptek Mini-X X-ray Tube), and the solar cell mounted in a dark light-tight chamber. (b) Semiconductor parameter analyzer (Keithley 4200A-SCS) used to measure solar cell I-V characteristics and short-circuit current response.

### C. Pulsed Laser Measurements

A pulsed x-ray source that produces fast switching x-ray pulses may be used to evaluate the transient response of solar cells. However, at the time of this study, such a facility remained inaccessible. Thus, a pulsed laser source with fast rising laser pulses was used as an alternative testing tool. Specifications of the laser source are summarized in Table II.

TABLE II. SPECIFICATIONS OF THE LASER SOURCE USED IN THIS STUDY

|                            |                                                                                  |
|----------------------------|----------------------------------------------------------------------------------|
| Laser Type                 | Solid-state diode pulsed laser                                                   |
| Manufacturer, Model        | Thorlabs, NPL41B                                                                 |
| Operational wavelength     | 405 nm                                                                           |
| Pulse width                | Variable from 6 ns to 38 ns                                                      |
| Pulse rise time            | 1 ns to 2 ns                                                                     |
| Pulse repetition frequency | 1 MHz, 5 MHz, 10 MHz fixed; 0 MHz to 10 MHz variable with external trigger input |

In these measurements, the solar cells were positioned in front of the laser source and illuminated by the laser beam. The short-circuit current resulting from the solar cell was amplified by a high-speed current amplifier into an equivalent voltage signal that was measured using an oscilloscope (Fig. 4(a), 4(b)).

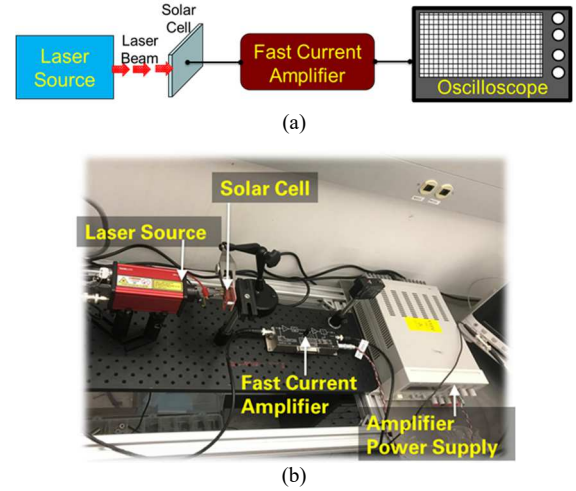


Fig. 4. (a) Schematic of the experimental setup used for measurement of solar cell response to fast switching nanosecond laser pulses (b) A photograph of the actual setup used in these measurements, showing the pulsed laser source, perovskite solar module and the fast current amplifier.

### D. Transient X-ray Response

For these measurements, a continuous beam of x-rays from the x-ray tube (see Table I) was transformed into x-ray pulses using a mechanical chopper. The chopper consisted of a rotating stainless-steel (SS) chopper blade as shown in Fig. 5(a). The thickness of the SS blade was about 760  $\mu\text{m}$ , sufficient to attenuate 99% of 30 keV or  $\sim 70\%$  of 50 keV photon flux. The output pulse from the chopper is a trapezoid, with flat top duration that depends on the difference in sizes of beam diameter and slot width (Fig. 5(b)). If the beam diameter is same as the slot width, the flat top duration is zero which gives rise to a triangular pulse. Due to the presence of 10 slots in the chopper blade, the trapezoidal x-ray pulse frequency is 10 times that of the chopper rotational frequency ( $f_c$ ).

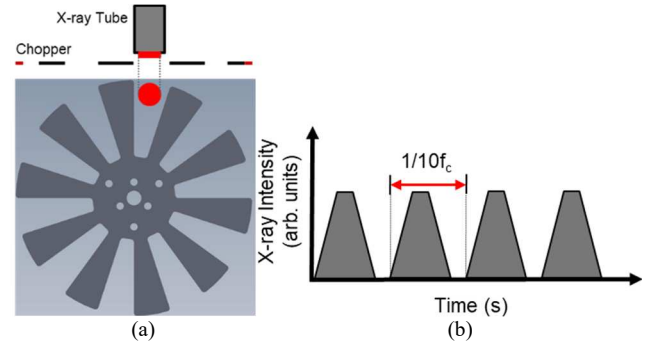


Fig. 5. (a) An illustration of the method used to produce pulsed x-rays, showing the x-ray tube and chopper blade in both top view and front view (b) A visualization of the x-ray pulse produced by the chopper setup, indicating the relationship between chopper rotational frequency and x-ray pulse frequency.

The limitation of this pulsed x-ray setup is that the chopper cannot produce fast rising pulses. As the rotational frequency is constrained to 100 Hz by the chopper controller, the x-ray pulse frequency and the rise time are limited to 1 kHz and 100s of microseconds. Nevertheless, the motivation for these measurements was to gain a valuable insight into the



evaluation of solar cell transient response to pulsed ionizing radiation.

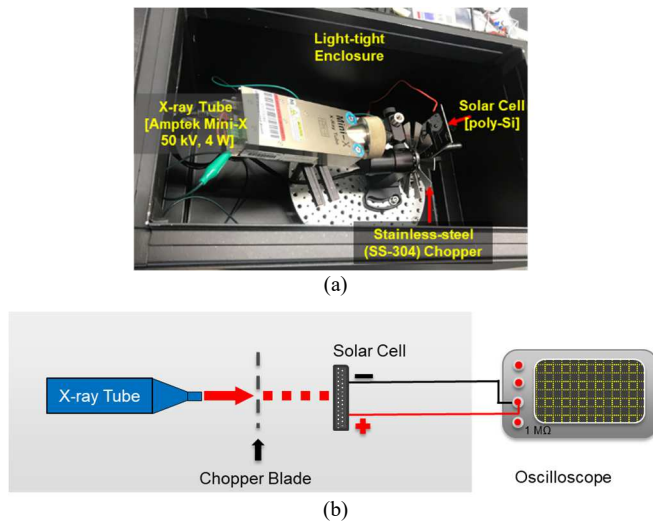


Fig. 6. (a) Pulsed x-ray source setup showing the continuous x-ray source (Amptek, Mini-X), SS chopper blade and solar cell device under test, mounted on an aluminum platform and enclosed in a metal housing (b) Schematic of the experimental setup used to measure the transient response of solar cell to x-rays.

The experimental setup used in pulsed x-ray measurements consisted of the Amptek Mini-X x-ray tube producing continuous x-rays, the SS chopper blade and the solar cell housed in a light-tight metal enclosure (Fig. 6(a)). During the tests, the enclosure was shielded on all sides using 6.35 mm lead sheets.

We measured the Si solar cell response to pulsed x-rays at different x-ray tube voltages corresponding to different x-ray energies. The output terminals of the solar cell were connected to a high impedance input of a mixed signal oscilloscope (Tektronix, MSO54) to measure the solar cell output voltage (Fig. 6(b)).

### E. 2-MeV Fast Neutron Response

In this step, we evaluated the solar cell response to fast neutrons from a thermal fission spectrum by utilizing a fast neutron beam facility at The Ohio State University Research Reactor (OSURR). A neutron chopper was designed to produce time-modulated neutron beam intensity and provide neutron pulses to evaluate the transient response of a solar cell. Fig. 7(a) shows a visualization of the experimental setup used in these measurements. The neutron chopper consists of a borated high-density polyethylene (BPE) cylinder placed at the exit of beam collimator with its axis of rotation perpendicular to the beamline. The BPE cylinder contains a rectangular aperture on its lateral surface, of depth equal to the cylinder diameter (= 12 inches). Neutrons are always blocked except when the aperture starts to align with the exit window of the neutron beam collimator. Neutrons, thus exiting the rectangular aperture on the chopper will have their maximum intensity when the axes of both the collimator and the aperture are aligned in parallel (Fig. 7(b)). Thus, the resulting neutron pulse would have a slow rise to its maximum and an equally slow fall to the baseline depending on the chopper rotational

frequency. This design exploits the cylinder diameter to provide a large attenuation length for neutrons, instead of the cylinder height that can be lowered enough to just cover the beam area.

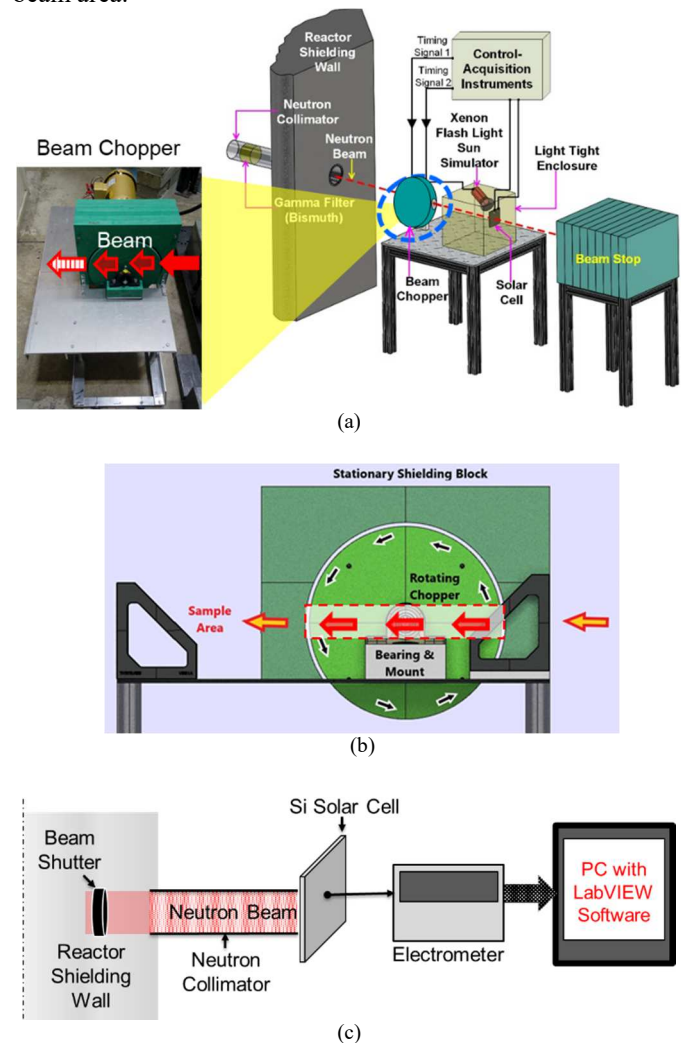


Fig. 7. (a) A visualization of the experimental setup used in the measurement of a solar cell response to time modulated neutron beam, showing the neutron beamline at the OSURR, beam chopper, and solar cell and flashlight in a light-tight enclosure, and data acquisition and control instruments. (b) A CAD drawing of the neutron beam chopper setup showing the direction of rotation of chopper cylinder with respect to the neutron beam path. (c) Schematic of the measurement setup used in the measurement of steady-state response of solar cell to neutrons.

At first, the solar cell's steady-state response to neutrons was evaluated as preliminary step to the study of transient response of solar cell to pulsed neutrons. The Si solar cell was placed at the collimator exit and exposed to a steady beam of neutrons with  $\sim 2$ -MeV flux on the order of  $10^7 \text{ cm}^{-2} \cdot \text{s}^{-1}$ . The solar cell output current was measured using an electrometer and recorded on a PC in real-time using LabVIEW software (Fig. 7(c)). The fast neutron beamline at OSURR has a beam shutter located inside the reactor shielding wall, which consists of a thick lead block. The beam shutter, when fully closed, blocks the gamma-ray component in the beam, while allowing the neutrons partially. The beam shutter was closed and opened in succession to observe the solar cell response to changes in the beam intensity.

### III. RESULTS AND DISCUSSION

#### A. Steady State X-ray Response

Results from steady-state x-ray measurements of a Si solar cell performed at different x-ray tube voltages (kV) are shown in Fig. 8. The kV value determines the maximum x-ray energy in the x-ray spectrum and also affects the radiation output of the x-ray tube, which is roughly proportional to  $kV^2$ . Thus, although a constant tube current (20  $\mu$ A) was used in each case, the resulting x-ray intensity increased as the kV value increased. The Si solar cell I-V characteristics show a proportional increase in both the short-circuit current ( $I_{sc}$ ) and open-circuit voltage with increasing x-ray energy (Fig. 8(a)). It is to be noted, however, that the  $I_{sc}$  of a solar cell is lower than the actual x-ray induced photocurrent ( $I_{ph}$ ), and depends on parameters such as the solar cell series and shunt resistance, and capacitance. The plot on the right shows a time-varying current response of Si solar cell to x-rays as a function of the kV value (Fig. 8(b)). In these measurements, the x-ray source was turned ON and OFF manually at different kV settings. The result shows a stable  $I_{sc}$  response of the solar cell, despite its thin depletion region, that accurately reproduced the variation in input x-ray energy and intensity as a function of time.

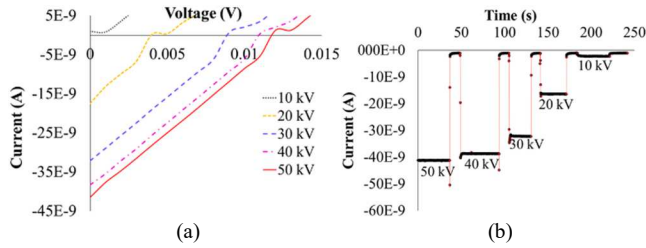


Fig. 8. (a) I-V characteristics of a commercial Si solar cell under x-ray illumination (Amptek, Mini-X X-ray tube) (b) Si solar cell  $I_{sc}$  vs. time under x-ray illumination at varying tube kV.

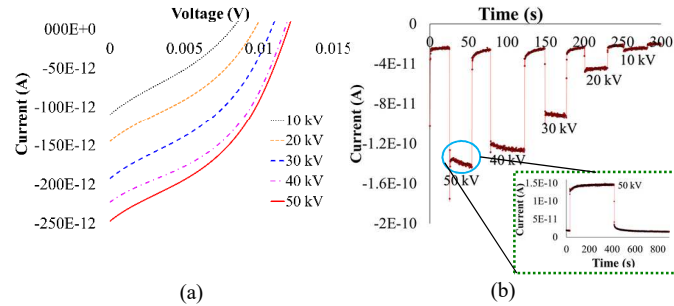


Fig. 9. (a) I-V characteristics of an in-house fabricated perovskite solar cell under x-ray illumination (Amptek, Mini-X X-ray tube) (b) Perovskite solar cell  $I_{sc}$  vs. time under x-ray illumination at varying tube kV.

The x-ray measurements were repeated for perovskite solar cells and the results are shown in Fig. 9(a), 9(b). The perovskite devices showed a similar behavior as that of Si solar cell under x-ray illumination. The  $I_{sc}$  is however, much lower than that of Si, which is explained by the small device area and thin active region of the perovskite solar cells. Another interesting feature is the slow rise to steady-state value and slower decay to baseline for each kV setting. An exemplary result is shown in the inset of Fig. 9 for  $I_{sc}$

measurement at 50 kV, which indicates the long duration to reach steady-state and longer decay. This behavior may be attributed to the effects of ion migration on electron-hole distribution and transport in the perovskite solar cell.

The above x-ray measurements, however, have not revealed the solar cell response to a fast-transient radiation, as the x-ray tube only produced a continuous beam. It is expected that a nuclear detonation would result in the sudden onset of radiation pulse at the location of a solar farm or panel located few kilometers away from the detonation point. In this context, the transient response of solar cells was first evaluated using pulsed laser radiation as discussed below.

#### B. Pulsed Laser Measurements

In these measurements, a commercial-off-the-shelf laser system (Table II) was used that produced nanoseconds duration laser pulses with fast rising edges. Both solar cell devices produced a fast transient response to the incoming laser pulses, which was exemplified by risetimes typically less than 1  $\mu$ s and decay times on the order of a few  $\mu$ s. Fig. 10 shows the transient signal from Si and perovskite solar cells in response to 20 ns laser pulses at 1 kHz pulse repetition frequency. The pulse in green color represents the 1 kHz external trigger signal supplied to the laser source, whereas the yellow colored pulse represents the output signal from solar cell. The transient signal showed up almost instantly at the arrival of laser trigger pulse with rise times of 61 ns and 194 ns for the perovskite and Si solar cells, respectively. These results demonstrate that the solar cells can detect a fast transient radiation and produce a proportional measurable output signature. It is to be noted that a fast current amplifier with low input impedance (50  $\Omega$ ) was used to convert the current output from solar cell to a voltage signal. Due to this small load resistance, the decay time constant of the solar cell current pulse and thus, the output voltage pulse is much smaller.

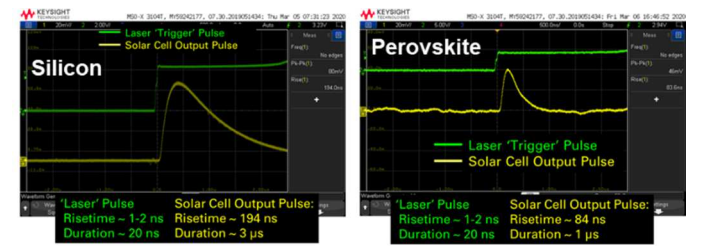


Fig. 10. Left: Si solar cell response to 405 nm pulsed laser illumination. Right: Perovskite solar cell response to 405 nm pulsed laser illumination

#### C. Transient X-ray Response

The transient response of the Si solar cell to pulsed x-rays was measured at different x-ray tube voltages and a constant tube current of 50  $\mu$ A. Fig. 11(a) shows the solar cell output voltage waveform at different x-ray tube kV and an x-ray pulse frequency of 20 Hz. The output voltage increases in proportion to the x-ray energy and follows the input x-ray pulses with same frequency.

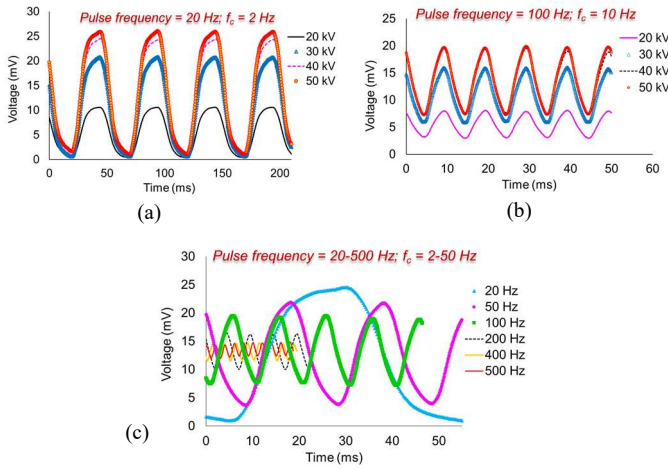


Fig. 11. (a) Output voltage waveform of the Si solar cell in response to x-ray pulses of 20 Hz pulse frequency at various tube kV and 50  $\mu$ A tube current. (b) Output voltage waveform of the Si solar cell in response to x-ray pulses of 100 Hz pulse frequency at various tube kV and 50  $\mu$ A tube current. (c) Output voltage waveform of the Si solar cell in response to x-ray pulses of varying frequency at 40 kV and 50  $\mu$ A tube voltage and current, respectively.

At an x-ray pulse frequency of 100 Hz, the Si solar cell exhibits similar voltage response with increasing x-ray energy (Fig. 11(b)). However, the pulse amplitude is lower compared to that at the 20 Hz frequency. To further understand the chopper frequency dependence of solar cell response, measurements were performed at different frequencies and the results are shown in Fig. 11(c) for x-ray pulse frequencies from 20 Hz to 500 Hz at 40 kV, 50  $\mu$ A tube setting. As the frequency increases, the amplitude of voltage waveform decreases, which is a result of the large solar cell  $RC$  time constant. The large  $RC$  constant is manifested in the slow output voltage response, where the output voltage has a settling time longer than the input pulse duration. Thus, as the input x-ray pulse duration decreases, amplitude of the solar cell output voltage decreases further. It is to be understood however, that, in the context of a nuclear detonation, the parameters of interest are the risetime and duration of a single radiation pulse and not the pulse repetition frequency.

#### D. 2-MeV Fast Neutron Response

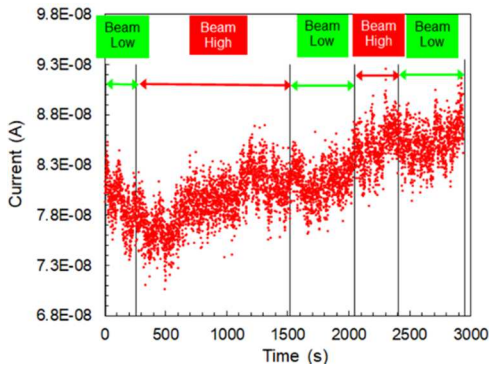


Fig. 12. Short-circuit current response of the Si solar cell to neutrons at steady-state as the neutron intensity is modulated intermittently using a beam shutter.

Fig. 12 shows the short-circuit current response of the Si solar cell to a steady neutron exposure at the fast neutron beamline of the OSURR. The solar cell response shows a rough correlation to the incident neutron intensity that was modulated manually using a beam shutter. The short-circuit current is seen to reproduce the neutron intensity pattern, with the current signal changing in proportion to the beam intensity. The response, however, is noisier and post the first cycle of shutter close-open, a slight deviation from the expected behavior is seen, which may be attributed to the contribution of delayed effects of neutron interaction i.e. neutron activation, in the solar cell. The contribution of residual gamma-rays in the neutron beam might not be trivial, which needs to be quantified.

#### IV. CONCLUSIONS

The measurements of solar cell output current and voltage using both steady-state and pulsed x-rays demonstrated good x-ray sensitivity of Si and perovskite solar cells. The pulsed laser measurements proved that, despite variation in response times, the solar cells are able to detect a fast transient radiation and produce proportional measurable output signature. The steady-state Si solar cell response to neutrons reproduced the changes in neutron intensity. Future work will focus on the study of transient response of the solar PV devices to time-modulated neutrons and gamma-rays using both simulations and experiments. Monte Carlo simulations will be performed to calculate charge generation and finite element simulations will be used to understand the charge transport in a solar cell and calculate its current response.

#### ACKNOWLEDGMENT

We would like to acknowledge the support of The Ohio State University Nuclear Reactor Laboratory and the assistance of the reactor staff members Susan White, Kevin Herminghuysen and Andrew Kauffman for the reactor neutron beam operations.

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